

Cracking the Olympic Medal Table

Summary

The flourishing development of the Olympic Games has attracted increasing public attention. Apart from individual events, the overall "medal table" of the country is undoubtedly the focus of our attention. Therefore, we aim to predict the Olympic medal table and analyze the key factors that determine national victory or defeat.

In Task 1: After data preprocessing, we use **Pearson's coefficients** and Statistical tests to extract factors that affect countries' Olympic medals, including the number of events participated, athlete awards, total historical medals, and host effect. We build a medal count prediction model based on **Multiple linear regression** and **Bayesian model** from a statistical perspective. Considering the number and type of events, we use the event **relaxation function** to correct the model. Making feasible assumptions about the data, we predict that 84 countries will make it to the medal table at Los Angeles, USA summer Olympics in 2028. The specific results and interval prediction are shown in Table 8. Among them, **Turkey** will increase the fastest predicted in 2028, while **Ireland** will do worse than in 2024.

In Task 2: We first extract the data features that affect the award rates of various countries, and analyze the impact of event selection on national award. Then, we build **K-Means** model based on the percentage of national awards and each event. Events can be divided into advantageous Events, balanced Events, and disadvantageous Events.

In Task 3: We use qualitative and quantitative analysis, **Wilcoxon signed Rank Test** and other statistical methods to verify the significant difference in medal count. We identify a significant correlation between coach replacement and medal count changes, especially in the US gymnastics and China volleyball events, showing that the introduction of coaches can improve athletes' competitive level significantly. We build an innovative quantitative model to evaluate the impact of great coach on the level of athletes and derive their contribution to the number of medals. In addition, we conduct analysis of countries such as **Ireland, Uzbekistan, and Iran**, identifying their performance in events and predicting the potential increase in medal count that results from the introduction of excellent coaches.

In Task 4: While analyzing the distribution of Olympic medals, we note that approximately 80% of the medals are amassed by just 20% of the athletes. This phenomenon indicates that there is a significant **Pareto effect**. Secondly, we have found that certain countries have performed exceptionally well in specific events, even achieving a complete monopoly on medals. In response to this phenomenon, we suggest that the Olympic Committees of various countries study the training methods and resource allocation of these successful countries to make improvements in the corresponding events. In addition, we have written a letter proposing suggestions to the National Olympic Committee for optimizing sports strategies.

Finally, we perform a sensitivity analysis of the model and investigate the effect of changes in the variable parameters of the model on the results.

Keywords: Multiple linear regression; Bayesian model; Wilcoxon signed Rank Test; K-Means;

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1 Introduction

1.1 Problem Background

The Olympic Games is one of the most influential sporting events globally. It not only showcases the outstanding talents of athletes but also promotes cultural exchanges and cooperation among nations. Since the revival of the modern Olympics in 1896, the Games has become an important platform for countries to display their athletic prowess. The medal table of the 2024 Paris Olympics shows the competitive landscape of countries in terms of gold medals and total medals. For example, the United States ranked first in total medals, while China and the United States were tied for first in gold medals. The host country, France, also performed well, ranking fourth. In addition, some countries such as Albania, Cabo Verde, Dominica, and Saint Lucia won medals for the first time in the Olympics, which reflected the diversity and inclusiveness of the Games.

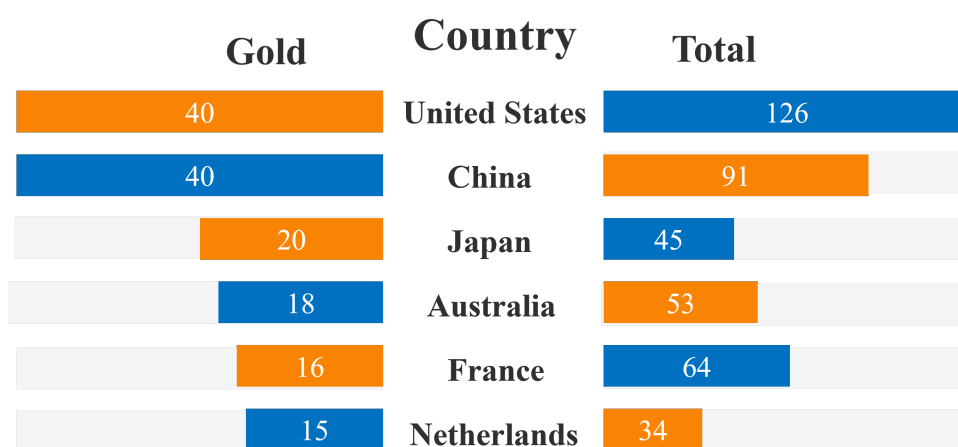


Figure 1: Paris Olympics(2024) Medal Table - Gold Medals Top 6 Countries

1.2 Restatement of the Problem

Considering the background information and the results in this file, we need to solve the following problems:

- ★ Develop a model to predict the number of medals that each country will win at the 2028 Los Angeles Olympics and provide prediction intervals. Predict how many countries may win medals for the first time in 2028, along with probability estimates.

- ★ Explore the relationship between the setting of Olympic events and the medal-winning situations of various countries. Identify which sports contribute the most to the medal counts of different countries and explain the reasons.

- ★ Identify the impact of "great coach" on the number of medals through data analysis. Select three countries and analyze which sports they require the introduction of "great coach", and estimate the potential impact.

- ★ Provide other original insights revealed by the model regarding the Olympic medal table. Explain how these insights can provide decision support for country Olympic Committees.

1.3 Our Work

The rest of this article is organized as follows. In the second section, we introduced the underlying assumptions and reasons, and in the third section, we mentioned common variables in the formula. In the fourth section, data preprocessing was performed before modeling. The fifth section established a prediction model for the number of medals and provided a prediction interval, exploring the importance of selecting sports events in different countries. The seventh and eighth sections continue to explore the interesting features of data files. In Sections 9 and 10, we analyzed the sensitivity of the model and further evaluated its strengths and weaknesses. Finally, based on our findings, a conclusion was drawn.

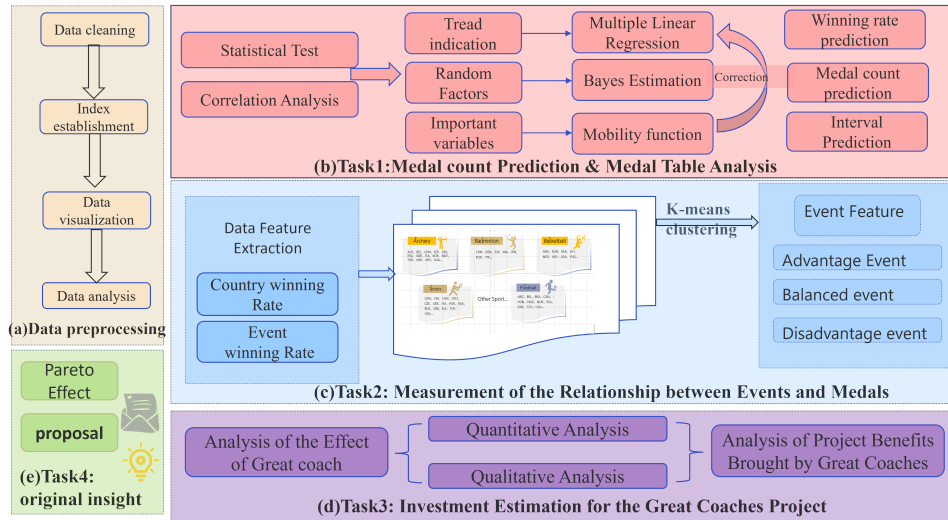


Figure 2: Our Work

2 Assumptions and Justification

To simplify the problem and make it convenient for us to simulate real-life conditions, we make the following basic assumptions, each of which is properly justified.

◆ **Assumption 1: The countries participating in the 2024 Olympics will continue to participate in the 2028 Olympics.**

Justification: Taking into account the core position of the Olympics in international sports exchange, the long-established tradition of participation by various countries, and the multi-faceted benefits brought about by participation in the Olympics, including political, economic, and cultural aspects, it is reasonable to assume that the countries participating in the 2024 Olympics will still choose to participate in the 2028 Summer Olympics in Los Angeles, USA.

◆ **Assumption 2: The medal performance of each country in different events is independent, meaning that the medal performance of a country in one event does not affect its performance in other events.**

Justification: The events in different competitions have varying project settings, participants, and competition rules. Therefore, a country's performance in one event will not directly affect its performance in other events. Thus, it is reasonable to assume that a country's medal performance in one event will not influence its performance in other events.

◆ **Assumption 3: Ignoring the influence of athletes' subjective factors, athletes partic-**

ipating in the Olympics are unlikely to withdraw from the competition midway due to a lack of subjective willingness to continue.

Justification: Due to undergoing multiple rounds of selection and long-term training, they are filled with a sense of mission and honor, and the event support and incentives are in place. This assumption can simplify the analysis of competition results and athlete performance, allowing for a focused exploration of the impact of objective factors.

3 Notations

Symbols	Description
$MP^i(t)$	The proportion of the total number of medals won by the i-th country at the t-th Olympic Games.
$S_Num^i(t)$	The total number of events participated by the i-th country in the t-th Olympic Games.
$P_Num^i(t)$	The number of awards won by athletes from the i-th country in the t-th Olympic Games.
$M_{i,j}(t)$	The number of awards won by country i in event j at the t-th Olympic Games.
$S_{i,j}(t)$	The total number of athletes from country i participating in event j at the t-th Olympic Games.
$\hat{MP}_j^i$	The j-th predicted value for country i obtained through a multiple linear regression model based on the independent variables.
$X^i(t), Y^i(t), Z^i(t)$	The number of gold, silver, and bronze medals won by country i in year t.
$Item_{i,j}(t)$	The index of advantage for country i in event j in year t.

where we define the main parameters while specific value of those parameters will be given later.

4 Data Pre-processing

4.1 Data Overview

Conduct a general analysis of the data as shown in Figure 3.

Notably:

- In 1936, during Nazi Germany, the Berlin Olympics was boycotted.
- In 1980, during the Cold War, the Moscow Olympics was boycotted by the West.
- In 1984, during the Cold War, the Soviet Union boycotted the Los Angeles Olympics in



Figure 3: Overview of four data tables

the United States.

These significant historical events caused large fluctuations in medal counts, resulting in some countries having zero medals, which can be analyzed further.

Encoding detection of the provided dataset files is shown in Table 1.

Table 1: Data files and encoding formats

File	Encoding
data_dictionary.csv	windows 1252
summerOly_athletes.csv	UTF-8
summerOly_hosts.csv	UTF-8 with BOM
summerOly_medal_counts.csv	UTF-8
summerOly_programs.csv	windows 1252

For convenience in subsequent processing, the above files were uniformly converted to .xlsx files in UTF-8 encoding format.

4.2 Data Cleaning

4.2.1 In "summerOly_athletes.csv" file

a) There is an athlete named "671" in row 244172, which is clearly not a valid name. To facilitate subsequent unified processing, after checking other information about the athlete, the name was corrected to Liu Qinglian.

b) Duplicate value check revealed 1466 duplicate rows in the file, which have been deduplicated.

c) The presence of an outlier of 0 in the total medal count for all countries in the following sports indicates that no country won any medals in those sports that year, which is clearly unreasonable. Therefore, these categories of sports should be removed.

	Year	Sport	yearly_sport_total_medal_num
719	2024	3x3 Basketball, Basketball	0
735	2024	Cycling Road, Cycling Mountain Bike	0
736	2024	Cycling Road, Cycling Track	0
737	2024	Cycling Road, Triathlon	0
748	2024	Marathon Swimming, Swimming	0

Figure 4: The statistical situation of the abnormal Sport category

4.2.2 In "summerOly_programs.csv" file

a) There are encoding issues, suspected to be due to encoding problems, so it was processed using the Windows 1252 encoding.

b) There are missing values in the programs.csv file. The detailed missing situation is shown in Figure 5.

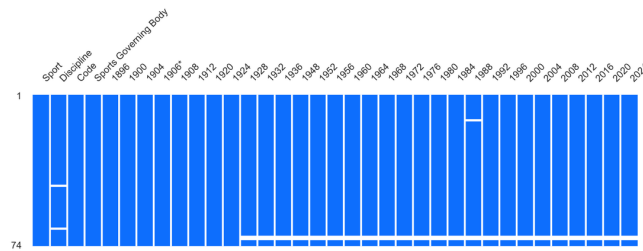


Figure 5: Schematic diagram of missing values in the programs table

i. The value for 1988 in row 14 is missing.

ii. Figure skating and ice hockey, which are ice sports, were held at the Summer Olympics before 1924. However, they were moved to the first Winter Olympics in 1924, so the values for figure skating and ice hockey in rows 71 and 72 for 1924 and later are missing.

iii. Because Modern Pentathlon and Water Motorsports do not have further classifications, the "Discipline" values are missing in rows 46 and 67. Based on other data analysis, the "Sport" values of the row were filled into the "Discipline" values.

4.2.3 In "summerOly_athletes.csv" and "summerOly_hosts.csv" files

a) Data for the year 1906 exist in "summerOly_athletes.csv" file but not in the hosts.csv file.

△ Reason: According to research, an unofficial Olympics was held in Athens, Greece, in 1906.

b) Data for the years 2028, 2032, 1944, 1940, 1916 exist in "summerOly_hosts.csv" file but not in "summerOly_athletes.csv" file.

△ Reason: The Olympics for the years 2028, 2032 have not yet taken place. The host countries and cities have been determined, but the participating athletes have not been determined. The three Olympics originally scheduled for the years 1944, 1940, 1916 were canceled due to

the two world wars in the 20th century. Although the hosts had been determined, there was no list of participating athletes because they could not be held.

4.2.4 In "summerOly_medal_counts.csv" file

After reading with UTF-8 encoding, it was found that the 'NOC' column had trailing spaces. All strings in this column were processed to remove excess leading and trailing spaces.

5 Task 1

5.1 Analysis of the Factors Influencing National Olympic Medal Counts from a Multidimensional Perspective

The number of Olympic medals won by a country reflects its overall strength and is influenced by multiple factors. Based on the existing dataset, we extract feature information to identify the important factors that affect the number of Olympic medals won by various countries. We then select significant features to analyze the quantity of Olympic medals.

It can be observed that the total number of medals won by each country is related to many factors, with the most significant influencing factors being the total number of events participated in by the country in the Olympics, the medal results of its athletes, the total number of medals won by the country in previous Olympic Games, and whether the country is the host nation.

(1) Total number of events participated in by the country in the Olympics: Analysis of the data reveals a significant positive correlation between the number of events participated in and the total number of medals. The more events participated in, the greater the competitive capacity of the country across various fields, thus increasing the chances of winning medals.

(2) Number of medals won by athletes: This article quantifies the competitive level of athletes based on their medal results in the events participated in. The more medals won by an athlete, the higher their competitive level is indicated to be. In the aforementioned correlation analysis, there is a significant correlation between the total number of medals won by the countries and this factor.

(3) Total number of medals won by the country in previous Olympic Games: Analyzing the medal data from past Olympic Games shows that the total number of medals has a certain degree of continuity and stability. Correlation analysis also indicates a significant correlation with this factor.

(4) Host nation effect: The host nation effect is a phenomenon widely studied and highly regarded in the competitive landscape of the Olympics. The correlation analysis above indicates that it has some influence on the number of medals won by the host country. Further examination through the Wilcoxon signed-rank test of the host advantage reveals that there is a significant difference between the number of medals won by the host nation and its previous medal count, further validating the impact of the host nation effect on the medal counts of countries.

5.2 Sub-model I: A Multivariate Linear Regression-Based Model for Predicting the Total Medal Count of Countries in the Olympics

5.2.1 Model Preparation

From the systematic analysis of influencing factors in the earlier stages, we know that the competitive performance of various countries in the Olympics, specifically quantified as the proportion of the total number of medals won in Olympic competitions, is closely related to the proportion of the total number of events in which the country participates in that year's Olympics. At the same time, the past performance of the country in the Olympics, especially in the previous year and recent years, has a significant and undeniable impact on the total number of medals for the current session.

Based on these factors, let $MP^i(t)$ represent the proportion of total medals won by the i -th country in the t -th Olympics; $S_Num^i(t)$ represents the total number of events participated in by the i -th country in the t -th Olympics; $P_Num^i(t)$ is the number of medals awarded to athletes from the i -th country in the t -th Olympic Games; $Home^i(t)$ is a dummy variable, when the i -th country is the host country of the t -th Olympics, $Home^i(t) = 1$, otherwise it equals 0.

5.2.2 Model Establishment

Because $MP^i(t)$ represents the medal-winning rate of each country, it takes values between 0 and 1. We perform a logarithmic transformation on $S_Num^i(t)$ and $P_Num^i(t)$, namely: $S_Num^i(t) = \log(S_Num^i(t) + 1)$, $P_Num^i(t) = \log(P_Num^i(t) + 1)$. Thus, it satisfies

$$MP^i(t, \theta) = a_1^i + a_2^i \times S_Num^i(t) + a_3^i \times P_Num^i(t) + a_4^i \times Home^i(t) + a_5^i \times MP^i(t-4) + \varepsilon^i(t) \quad (1)$$

Where, $\theta = [a_1, a_2, a_3, a_4, a_5]$ is the regression coefficient; t is the year of hosting the Olympics; $\varepsilon^i(t)$ encompasses all uncertain factors.

Then we fit it using the least squares method. Let $y^i(t)$ be the observed value for the i -th country in the t -year Olympics. The regression result is $\hat{MP}^i(t, \hat{\theta}) = \sum_{j=1}^5 a_j^i \times f^i(t, j)$, and the loss function is $L(\hat{\theta}) = \sum_t [y^i(t) - \hat{MP}^i(t, \hat{\theta})]^2$. The parameter estimate that minimizes $\hat{\theta}$ is taken as $L(\hat{\theta})$, thus $\hat{MP}^i(t, \hat{\theta})$ is the regression outcome.

The total number of medals at each Olympic Games is not constant; it is determined by a combination of factors such as the events contested and the athletes' conditions. The changes in the total number of medals at each Olympic Games over time are illustrated in Figure 6:

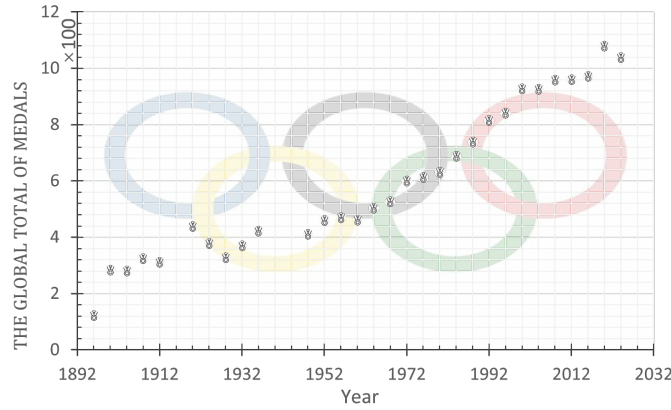


Figure 6: Global Total Olympic Medals Over Time

From the analysis of the above figure, it can be seen that the total count of each Olympic Games shows a linear trend over time. It is noted that data points for the years 1916, 1940, and 1944 are missing, as during these years the Olympics were cancelled due to historical events, thus no records exist. The scatter plot above indicates a generally linear trend. Therefore, we employ a linear function to fit this trend.

$$M_Sum = \beta_0 + \beta_1 t \quad (2)$$

Where, $\hat{\beta} = [\beta_0, \beta_1]$ is the regression coefficient; t refers to the year of the Olympics -1892 .

We employ OLSE to estimate $\hat{\beta}$, and predict that the total number of medals for the years 2028 and 2032 in Table 2.

Table 2: The prediction of the total number of medals.

Year	2028	2032
Pre_Sum	1026	1052

Combining the medal-winning rates of various countries with the predictions of the total number of Olympic medals, $Me^i(t) = Sum_M(t) \times MP^i(t)$, based on $Me^i(t)$, we can predict the standings on the medal table.

5.3 Sub-model II: Exploring the Medal Distribution of Countries Based on Bayesian Models

5.3.1 Model Preparation

At the early stages of the competition, we cannot ascertain the true capabilities of participating athletes, thus we can only predict that the probabilities of these four outcomes for each athlete are equal. Similarly, within the total medals awarded to each country, we use a multinomial distribution as the prior distribution, that is

$$(X, Y, Z|N) \sim Multinomial(N, p_1, p_2, p_3) \quad (3)$$

Where, p_1 , p_2 , and p_3 are the prior probabilities of gold, silver, and bronze medals, respectively; and $p_1 + p_2 + p_3 = 1$.

5.3.2 Model Establishment

Based on the specific medal counts of various countries, $D^i(t) = X^i(t), Y^i(t), Z^i(t)$ represents the number of gold, silver, and bronze medals country i wins in t year.

Here, $i \in \{France, UnitedStates, Germany, Australia, \dots\}$, $t \in \{1896, 1900, \dots, 2024\}$.

Assuming the observations for year t from country i are independent and identically distributed given the winning situation, the sample likelihood function can be expressed as $P(D^i|X^i, Y^i, Z^i) = \prod_t (p(x^i(t), y^i(t), z^i(t)|X, Y, Z))$.

Substituting the prior distribution and the likelihood function into the equation, and computing the normalization constant C ensures that $\sum_{x+y+z=N} P(X=x, Y=y, Z=z|D) = 1$.

Thus, the posterior distribution can be expressed as: $P(X^i(t), Y^i(t), Z^i(t)|D) = \frac{1}{C} \times \frac{N!}{x^i(t)!y^i(t)!z^i(t)!} P_1^x P_2^y P_3^z \prod_{t=1}^n \frac{N!}{x^i(t)!y^i(t)!z^i(t)!} P_1^{x^i(t)} P_2^{y^i(t)} P_3^{z^i(t)}$.

By introducing the Bayesian model, in combination with the prior distribution and historical data, we can effectively evaluate the number of gold medals won given the total number of medals is known.

5.4 Model Improvement Based on Advantage Project Relaxation Function

As the medal count does not adequately measure the ability of athletes from countries that have never won any medals, we modify the multiple linear regression model. In this case, the assessment of athletes' capabilities from a country is not only based on the total count of medals won but also considers all data regarding the events and awards of athletes provided in the data set, including results from unofficial competitions. Thus, we adjust the independent variable $P_Num^i(t)$ to ensure $P_1_Num^i(t) = P + P_Num^i(t)$, where P indicates the minimum level of the country's athletes, measured by the participation rate in events from that country. Additionally, the effect of project advantages also influences the medal-winning outcomes of countries to some extent. By considering the number and types of events, we adjust the independent variable $S_Num^i(t)$ as follows:

$$S_1_Num^i(t) = S_Num^i(t) + f(p) \quad (4)$$

Where, $f(p) = \sum_{k=1}^n w_k I_k$ and w_k represent the weight proportions for various projects; I_k represents the count variable for project types.

Therefore, the modified multiple linear regression function is $MP^i(t, \theta) = a_1^i + a_2^i \times S_1_Num^i(t) + a_3^i \times P_1_Num^i(t) + a_4^i \times H^i(t) + a_5^i MP^i(t-4)$.

Since the previously mentioned random variables $X^i(t)$, $Y^i(t)$, and $Z^i(t)$ do not constitute a completely independent variation process in practice, their random fluctuations and magnitudes are influenced by the types and number of events that the country participates in, thus the values of $P_1^i(t)$, $P_2^i(t)$, and $P_3^i(t)$ will change with the number and type of events, as categorized below:

$$Item_{i,j}(t) = \frac{M_{i,j}(t)}{S_{i,j}(t)} \times \frac{P_Num^i(t)}{S_Num^i(t)} \quad (5)$$

Where, $M_{i,j}(t)$ represents the number of medals won by country i in event j at the Olympic Games in year t ; $S_{i,j}(t)$ represents the total number of athletes from country i participating in event j at the Olympic Games in year t . The indicators can be defined as follows:

(1) Advantage Projects: The country has a strong capability in this event, with high athlete skill levels, a comprehensive training system, and a history of good performances in major competitions, resulting in a high probability of winning medals. Specifically, when $Item_{i,j}(t) \in (0.6, 1)$, it indicates that this event has long been a superior event for the country.

(2) Balanced Projects: The country has an average capability in this event, with a gap compared to the top levels, yet possesses certain competitive strengths, offering a possibility of winning but with some instability. Specifically, when $Item_{i,j}(t) \in [0.4, 0.6]$, it indicates that this event has long been a balanced event for the country.

(3) Disadvantage Projects: The country is lagging in this event, with athletes' levels, training conditions, and competitive experience far behind other countries, making it very difficult to win medals. Specifically, when $Item_{i,j}(t) \in (0, 0.4)$, it indicates that this event has long been an inferior event for the country.



Figure 7: Advantageous projects and corresponding countries

We observe that when a project is an advantage project for the country, there is a higher likelihood of winning gold medals, while performance in non-advantage projects is less optimistic. Therefore, we introduce an advantage project relaxation function $f(M)$ to correct the Bayesian model $P_j^{oi}(t) = P_j^i(t)f(M)$.

Therefore, the revised Bayesian estimate replaces P_1 with P_1^o , and by applying the Newton iterative algorithm, we can obtain the estimated value of P_1^o , thus predicting the number of gold medals each country will win as $Me^i \times P_1^{oi}$.

Table 3: The prediction of the award situations in 2028

Top 10			Bottom 10		
Country	rank	Medal	Country	rank	Medal
UnitedStates	1	106	Hungary	11	23
China	2	87	Cuba	12	22
GreatBritain	3	82	Canada	13	20
Germany	4	59	Ukraine	14	19
Australia	5	47	Brazil	15	18
Italy	6	42	Spain	16	18
Japan	7	42	Sweden	17	16
SouthKorea	8	35	Kenya	18	15
Netherlands	9	27	Norway	19	15
France	10	23	Poland	20	15

5.5 Uncertainty analysis of the influence model

The dataset we possess and the selected data features may not adequately represent all the factors influencing a country's medal wins, which could lead to some bias in our model. While multiple linear regression categorizes uncertainties as a random error term, the estimation of this random error term also carries bias. The limitations of the data we face when studying the factors affecting a country's performance in Olympic Games and similar events cannot be overlooked. The current dataset we have has a limited source and cannot comprehensively cover all potential influencing factors. Additionally, the chosen data features are based solely on existing knowledge and analytical frameworks, reflecting only a portion of the factors related to winning medals, making it difficult to fully represent all the factors that influence a country's medal wins.

In multiple linear regression, the residual e^i is the difference between the observed value MP_j^i and the model's predicted value $\hat{MP}_j^i$, calculated using the following formula:

$$e_j^i = MP_j^i - \hat{MP}_j^i \quad (6)$$

Where, MP_j^i represents the j -th observation of country i ; $\hat{MP}_j^i$ is the predicted value for country i 's j -th observation calculated by the multiple linear regression model based on the independent variables.

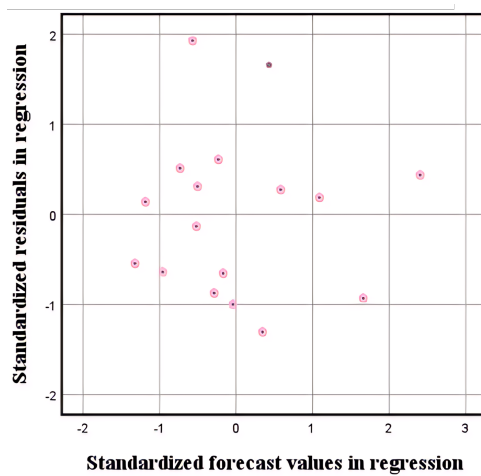


Figure 8: Residual plot

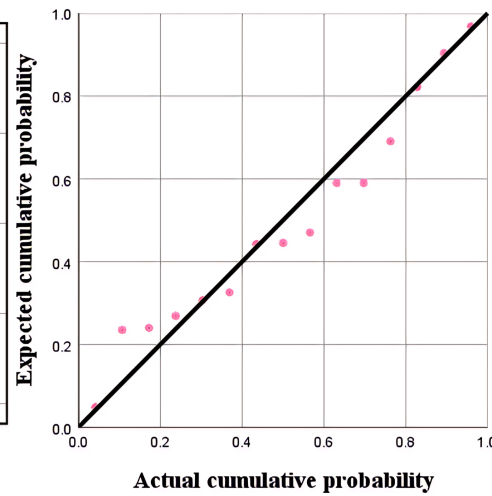


Figure 9: Quantile-Quantile plot

From the above residual plot and Quantile-Quantile plot, we can see that the residuals are evenly distributed around the horizontal axis, indicating that there is no heteroscedasticity. Additionally, the Quantile-Quantile plot is mostly close to the straight line, which implies that the predictions regarding cumulative probabilities made by this model are fairly accurate overall. This means that the model is capable of capturing the distribution characteristics of the data to some extent, demonstrating a degree of reasonableness and effectiveness.

5.6 Revealing the results of the medal table for future Olympic Games

5.6.1 Analysis of Prediction Model Results

Based on the modified multiple linear regression trend prediction model, we have obtained the medal-winning trends of participating countries since 1986, excluding those countries that

did not hold the Olympics normally, including countries that have never won a medal. The regression results for all participating countries have been obtained, and we will now list the regression results for some countries in the table below:

The estimated regression coefficients shown in the above table reflect the number and types of events in which each country participates, the overall performance of athletes in each country measured by the number of medals won, as well as a quantitative analysis of the past winning situations and the host country effect based on historical data. The regression results indicate that these factors positively influence the medal-winning rates of the countries. Next, we estimated the probability of each country winning gold medals using a modified Bayesian model, continually refining the probabilities with existing data to achieve optimal estimation results.

Based on historical data, assuming that the number and types of events as well as the performance levels of participating athletes in each country remain stable, and knowing that the host country is the United States with past winning situations for 2024, we predict the medal tally for the 2028 Los Angeles Summer Olympics. Additionally, we provide the prediction intervals with a confidence level of 0.95 for all countries, as shown in Table 5:

From the changes in the number of medals won by various countries, the ten countries with the most significant changes are the United Kingdom, Norway, New Zealand, the Netherlands, Iran, Canada, Uzbekistan, Cuba, Germany, and France. The changing trends are illustrated in Figure 10:



Figure 10: Chart of the Medal Counts Changes of the Top 10 Countries

Analyzing the figure, it can be concluded that the countries predicted to see significant improvements in their medal counts include Germany, Cuba, Norway, and the United Kingdom. Conversely, the countries with declining results should not be overlooked, notably France,

Table 4: Regression coefficient of the trend prediction model

Country	θ					R^2
	a1	a2	a3	a4	a5	
Australia	0.023	0.141	0.190	0.351	0.281	0.877
Belarus	0.011	0.054	0.113	0.000	0.764	0.769
Belgium	0.124	0.314	0.215	0.241	0.465	0.864
China	0.141	0.521	0.121	0.327	0.684	0.880
United States	0.118	0.381	0.241	0.261	0.763	0.792

Table 5: Prediction interval for the number of Medals in difference Country

Country	Pre_2028	Confidence interval of 0.95	
		LMCI_2	UMCI_2
United States	106	94	134
China	87	72	102
GreatBritain	82	41	123
Germany	59	37	80
Australia	47	37	57
Italy	42	30	55
Japan	42	33	51
SouthKorea	35	22	48
Netherlands	27	19	36
France	23	-4	51
Hungary	23	15	32
Cuba	22	13	32
Canada	20	12	28
Ukraine	19	4	33
Brazil	18	15	22
Spain	18	5	31
Sweden	16	7	26
Kenya	15	9	21
Norway	15	8	21
Poland	15	8	22

Uzbekistan, Canada, Iran, the Netherlands, and New Zealand. Specifically, compared to 2024, it is expected that France will see a reduction of 41 medals in 2028, which warrants attention.

Regarding the changes in rankings among countries, the ten countries with the most significant ranking changes are the Philippines, Cuba, Poland, Slovenia, Austria, Iran, Kyrgyzstan, Turkey, Uzbekistan, and Ireland. The changing trends are illustrated in Figure 11:

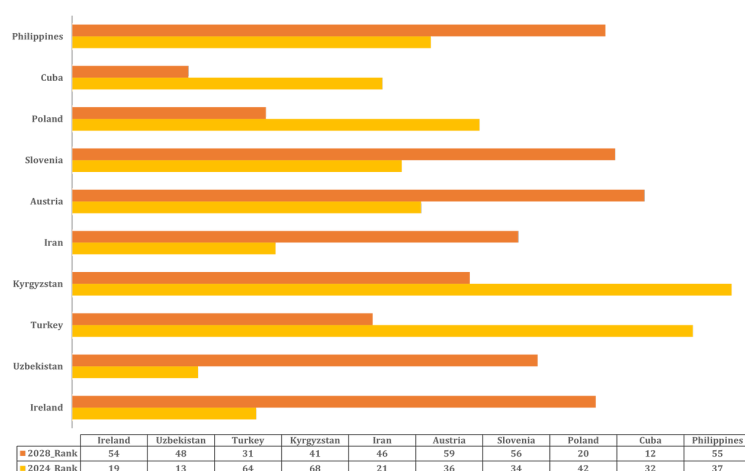


Figure 11: Chart for Comparing the Ranking Changes of the Top 10 Countries

Analyzing the figure, it can be observed that the countries with significant improvements in

their rankings include Turkey, Kyrgyzstan, Poland, and Cuba. Furthermore, it is predicted that the rankings of the Philippines, Slovenia, Austria, Iran, Uzbekistan, and Ireland will decline significantly. Notably, both Ireland and Uzbekistan are projected to drop 35 positions, marking the most pronounced changes in rankings.

The calculations regarding the likelihood of countries that have never won a medal obtaining their first medal at the 2028 Olympics, along with the corresponding estimated probabilities, are shown in Table 6.

After processing and analyzing the data, it was found that since the first Olympic Games in 1886, a total of 101 countries have never won a medal based on historical participation data. Among these countries, some are temporary teams, some have ceased to exist due to historical reasons, and others have failed to participate continuously due to special circumstances. This paper does not consider these special situations or distinguish among these countries, but rather focuses on their participation records and medal-winning situations.

Assuming these countries continue to participate in the 2028 Olympics, we used a multiple linear regression model to predict their medal-winning outcomes. The results indicate that 22 countries are likely to achieve a “breakthrough” in winning their first medal in history during the 2028 Olympics. However, the predicted probabilities of winning vary among the countries and are relatively low. This is largely due to the analysis of their past performance at previous Olympics and the lack of prominent competitive advantages in their events.

Considering the wide range of events included in the Olympics, which not only have numerous categories but also encompass everything from physically demanding sports like athletics and swimming to skill-based events like gymnastics and diving, as well as team sports such as basketball and football, an in-depth analysis of medal predictions reveals significant disparities in strength across different Olympic disciplines.

For instance, the United States showcases an absolute advantage in many events. Given this strong performance in favored events, continuing to invest in these areas is undoubtedly a key strategy to maintain its leading position in the Olympic medal table. However, for more balanced events, such as badminton, where the overall strength of the U.S. still has room for improvement compared to traditional strong teams in Asia, more investment should be made to elevate performance. In disadvantageous events, such as archery, the significant gap between the

Table 6: Table of the Countries Achieving Zero-breakthrough and Probabilities

Country	Probability	Country	Probability
Nicaragua	0.129217	Tuvalu	0.131417
Libya	0.112495	South Sudan	0.118437
Comoros	0.144345	Lebanon	0.131196
Malawi	0.130465	Democratic Republic of the Congo	0.145907
Rhodesia	0.131206	East Timor	0.102365
Cook Islands	0.100384	Eswatini	0.097714
Peri	0.137529	St Kitts and Nevis	0.116556
South Vietnam	0.140657	StVincentGrenadines	0.122315
Cambodia	0.148522	Timor-Leste	0.13819
Kingfisher	0.117098	Brunei Darussalam	0.097726
Yangon	0.129572	DR Congo	0.120408

United States and archery powerhouses like Korea and China indicates the need for continuous efforts to enhance capabilities and overcome the disadvantages.

6 Task 2

6.1 Great Coach Effect

The Great Coach Effect refers to the positive impact that an outstanding coach has on a team or individual athletes. This effect encompasses various aspects, including improved athlete performance, enhanced team cohesion, and increased personal skills and tactical understanding. When Lang Ping coached the Chinese women's volleyball team, she created opportunities for each player to showcase their potential and unleash their energy. Bela Karolyi not only focused on improving athletes' skills but also emphasized the cultivation of their mental resilience, ensuring they could perform consistently well in critical moments.

Through the review and study of literature, we found that coaches play a crucial role in enhancing athletes' performance levels. According to the analysis results from Model One, the level of athletes is a core factor determining a country's medal-winning outcomes at the Olympics. It is evident that coaches have a direct influence on athlete performance, which in turn indirectly affects a country's Olympic performance—specifically, the number of medals won. To further explore the impact of the Great Coach Effect on the number of medals, we will focus on the direct effect as our entry point. First, we will analyze the direct influence of coaches on athlete performance. After clarifying this influence mechanism, we will use the model established in the first question to further deduce the impact of coaches on their country's medal count. This research approach will help us to gain a more comprehensive and in-depth understanding of the key role coaches play in their nation's Olympic achievements.

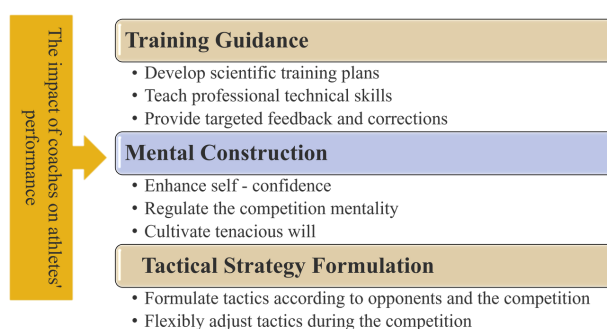


Figure 12: The impact of coaches on athletes' performance

6.2 Qualitative Analysis

The Károlyi gym, established by Márta Károlyi in the late 1980s, has become an exceptional training facility in the United States. Following this development, we can observe a significant increase in the average number of medals won by the U.S. women's gymnastics team, with 1980 serving as a pivotal dividing line.

The transition from having no gold medals to winning them can be clearly observed in the chart below. Prior to 1980, the United States had not won any Olympic gold medals in women's gymnastics. However, after 1980, the number of gold medals won increased significantly, not just by one or two, but rather by a remarkable surge.

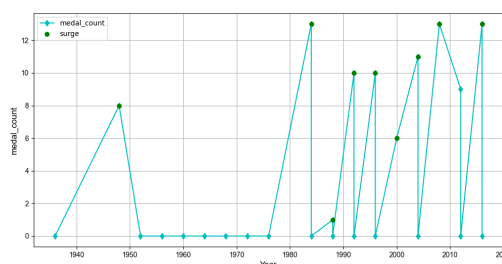


Figure 13: The number of Olympic medals won by the United States in women's gymnastics

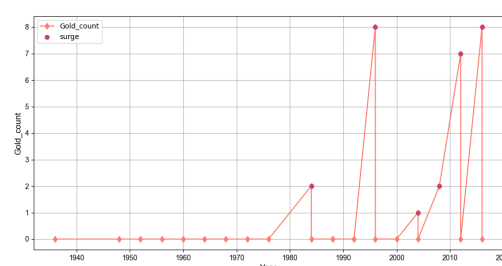


Figure 14: The number of Olympic gold medals won by the United States in women's gymnastics

By comparing the historical medal counts for women's volleyball in China and the United States, we can identify two key focal points in time. The first is 1995, when Lang Ping became the head coach of the Chinese national team, leading them from a period of decline to success. The second point is in 2008, when Lang Ping took over as head coach of the U.S. team, which had gone three consecutive years without winning any medals. Under her guidance, the team not only secured medals but also maintained their success for two consecutive years.

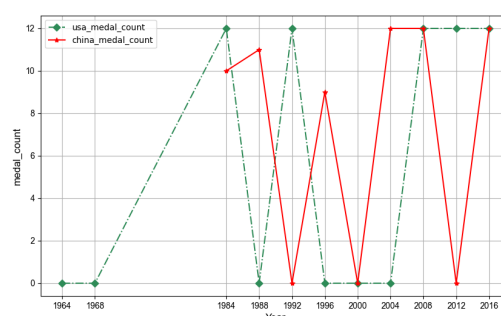


Figure 15: The number of Olympic medals won by China and the United States in women's volleyball

To further analyze the impact of outstanding coaching effects on the number of medals, a one-sample Wilcoxon signed-rank test was conducted on the above data.

Based on the variable Medal_count in U.S. gymnastics and its mean of 1.12, the significance P-value is 0.015, which is less than 0.05, indicating statistical significance. Therefore, we reject the null hypothesis, suggesting that there is a significant difference in award outcomes before and after the introduction of coaching. Similarly, for other sports, there are also significant differences in award outcomes before and after the introduction of outstanding coaches.

6.3 Quantitative analysis

First, we quantify the performance level of athletes from this country by calculating the medal-winning rate for all athletes in their respective events. The formula for this calculation is:

$$P_{i,j}(t) = M_{i,j}(t)/S_{i,j}(t) \quad (7)$$

Where, $P_{i,j}(t)$ represents the medal-winning rate of country i in event j at the Olympic Games in year t ; $M_{i,j}(t)$ indicates the number of medals won by country i in event j at the Olympic Games in year t (counted by the number of medals, not by events); and $S_{i,j}(t)$ represents the total number of athletes from country i participating in event j at the Olympic Games in year t .

Then, $\Delta P_{i,j} = P_{i,j}(t_1) - P_{i,j}(t_0)$ represents the change in the performance level of country i 's athletes in event j from the Olympic Games in year t_0 to the Olympic Games in year t_1 . Furthermore, assuming that the coach started coaching in a certain year t_0 , we can quantify the impact of the coach on athletes' performance levels by comparing the average changes in athlete performance level before and after the coach's arrival:

$$\Delta P_{i,j} = \bar{P}_{i,j}^{after}(t_0) - \bar{P}_{i,j}^{before}(t_0) \quad (8)$$

Where, $\bar{P}_{i,j}^{after}(t_0)$ represents the average performance level of country i 's athletes in event j after the coach's arrival; $\bar{P}_{i,j}^{before}(t_0)$ represents the average performance level of country i 's athletes in event j before the coach's arrival. Thus, the impact of the coach on the total number of medals can be expressed as:

$$\Delta M_{i,j}(t) = \Delta P_{i,j} \times A_{i,j}(t) \quad (9)$$

Where, $\Delta M_{i,j}(t)$ represents the number of additional medals won by country i in event j at the Olympic Games in year t after the coach's arrival.

Based on the available data, it can be calculated that for the United States in gymnastics, the quantified performance level of athletes before Márta Károlyi's coaching was 0.3125. After he became the coach, the athletes' performance level clearly improved in the subsequent Olympic Games, reaching 0.751. Similarly, for China in volleyball, before Lang Ping's coaching, the team had been at a disadvantage, with a quantified athlete performance level of only 0.1756. After Lang Ping took over as coach in 1995, China's quantified athlete performance level increased to 0.751. This further validates that the influence of coaches on athlete performance levels is profound.

6.4 Investment in Excellent Coaches Based on Benefit Analysis

The impact and significance of the Olympics on various countries are profound, serving not only as a key window to showcase athletic strength but also as a powerful booster for enhancing international image. Therefore, investing in great coaching programs holds significant and irreplaceable value, yielding multifaceted returns. Through scientific training methods, excellent coaches are always able to significantly enhance the overall performance of athletes and teams, thereby bringing high returns and efficiencies to their countries.

Based on the analysis of the medal standings for the United States at the 2028 Los Angeles Summer Olympics in the first question, an analysis of the medal outcomes for the 2024-2028 Olympics revealed that Ireland, Uzbekistan, and Iran ranked high in the 2024 medal standings. However, their predicted performances in the 2028 standings do not look optimistic. This prediction undoubtedly serves as a wake-up call for these three countries. Given the multiple profound impacts of the Olympics on nations, it is highly probable that these countries will take a series of strong measures to enhance their athletes' competitive levels.

Therefore, this paper chooses these three countries, which ranked among the top yet are predicted to see a significant decline in performance compared to 2024, for analysis. Further analysis of their respective strong, balanced, and weak events is detailed in the table below:

Through the mining of the existing dataset, it is found that for Ireland, the strong events are Aquatics, Athletics, and Gymnastics, whereas Boxing and Rowing are the weak events; other events can be considered the balanced events for the country. For Uzbekistan, the strong events are Boxing, Taekwondo, and Weightlifting, while Gymnastics and Judo are the weak events. For Iran, the strong events are Karate, Shooting, and Taekwondo, while Athletics and Weightlifting are identified as the weak events.

From the quantitative analysis results of the effects of excellent coaches, it is evident that after the introduction of excellent coaches, the competitive level of athletes in these events will improve to a certain extent, denoted as $\Delta P_{i,j}$.

$$MP^i = a_1^i + a_2^i \times S_{1_Num}^i + a_3^i \times P_{1_Num}^i + a_5^i M \quad (10)$$

Where, $P_{1_Num}^i = P_{ij} + \Delta P_{i,j} + P_Num^i$; M represents the historical number of gold medals won.

We use the following formula to estimate the impact of investment in the great coaching program on the increase in the number of medals won:

$$\Delta M = M_c - M_b \quad (11)$$

Where, ΔM represents the additional number of medals obtained after the introduction of excellent coaches; M_c refers to the predicted number of medals won after the introduction of excellent coaches; and M_b denotes the number of medals won without the introduction of excellent coaches.

Thus, ΔM is the increase in the number of medals attributable to the investment in the excellent coaching program for these three countries.

7 Task 3

7.1 Preliminary Analysis

In the early years of the Olympics, the participation rate of women was extremely low. However, in recent years, with the advancement of civilization, women have gradually begun to emerge in the Olympic Games. The proportion of female participants in the total participation records has steadily increased, and by the 2024 Olympics, the participation rate of women is expected to be on par with that of men.

The majority of participants did not win any medals: 78.67% of individuals (102,269 people) did not receive any medals. This observation reveals a clear Pareto effect, where 80% of the medals are concentrated among 20% of the participants, indicating that the vast majority of Olympic participants did not secure any medals.

A minority of individuals won medals: only 16.12% of participants (20,957 people) earned one medal, while 3.50% (4,546 people) won two medals. This demonstrates that the number of medal winners is relatively small, and the count decreases rapidly as the number of medals increases.

Sparsity of medal distribution: As the number of medals increases, the number of individuals who win medals decreases significantly. Only 0.14% of participants (187 people) won five medals, and 0.09% (113 people) won six medals, with the number of winners declining sharply after three medals.

A very small number of individuals achieved a high medal count: the data indicates that the number of participants who won ten or more medals is exceedingly rare, with only 17 individuals winning ten medals, and just one person winning 18 and 28 medals, respectively. This suggests that instances of winning a high number of medals are extremely uncommon within this dataset.

Distribution skewness: Overall, this distribution exhibits a pronounced positive skewness, indicating that the majority of individuals cluster around low medal counts (0-2 medals), while the distribution of higher medal counts is relatively sparse.

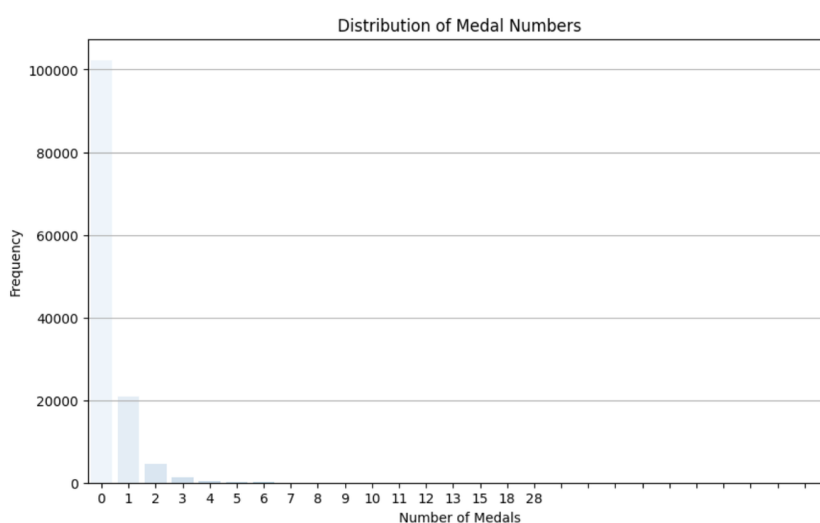


Figure 16: The changing participation rate of female athletes in various Olympic Games

We also observe that certain events are dominated by specific countries in terms of medal acquisition. For instance, countries such as the USA (Roque), SUI (Aeronautics), FRA (Croquet), GBR (Racquets), and ESP (Basque Pelota) have achieved a 100% medal share in their respective sports events, indicating that these nations possess an absolute advantage in these particular competitions.

Table 7: A descending list of the percentage of medals relative to the total number of medals

NOC	Sport	medal_num	total_medal_num	percentage
USA	Roque	3	3	100
SUI	Aeronautics	1	1	100
FRA	Croquet	8	8	100
GBR	Racquets	10	10	100
ESP	Basque Pelota	2	2	100
GBR	Cricket	21	24	87.5
GBR	Motorboating	6	7	85.71429
USA	Golf	41	58	70.68966
GBR	Jeu De Paume	2	3	66.66667
CAN	Lacrosse	36	60	60
GER	Alpinism	2	4	50
SUI	Alpinism	2	4	50
GBR	Polo	30	65	46.15385
CHN	Table Tennis	104	228	45.61404
CHN	Trampoline Gym	5	12	41.66667
USA	Ice Hockey	11	27	40.74074
CHN	Badminton	83	216	38.42593
JPN	Skateboarding	9	24	37.5
GBR	Figure Skating	10	27	37.03704
CHN	Trampolining	11	30	36.66667
SUI	Cycling Mountain Bike	4	11	36.36364

Additionally, there is a notable number of countries and events with moderate medal shares. For instance, CAN (Lacrosse) and GER, SUI (Alpinism) have medal shares ranging from 50% to 60%, demonstrating a certain level of competitiveness, although they do not reach a monopolistic status. Similarly, CHN (Table Tennis) and CHN (Badminton) account for 45.61% and 38.43% of the medals, respectively, indicating that China also performs strongly in these events.

From the above figure, it is evident that the diversification of sports events has a significant positive correlation with the number of Olympic medals obtained. To assess the relationship between the number of medals won by a country and the variety of sports events in which it participates, we employed correlation analysis. The results indicate that the Pearson correlation coefficient between the number of medals and the number of sports events participated in by the country is 0.6228, suggesting a strong positive correlation between the two variables.

7.2 Letter

Enjoy Your Bath Time!

Subject: Recommendations for Optimizing National Sports Strategy

Dear Leaders of the National Olympic Committee:

Following an in-depth analysis of the performance of various countries in international sports events, we have identified several key patterns and data that can provide important guidance for our future sports strategy.

Our analysis of medal distribution characteristics reveals that the vast majority of participants did not win any medals, with 78.67% of individuals failing to secure a medal, and 80% of the medals being concentrated among 20% of the participants. This phenomenon suggests that we should implement measures in two key areas: first, we must acknowledge the Pareto effect and focus on providing enhanced support and resources to those athletes who have already won medals, thereby helping them to achieve continued success in future competitions. Second, we should emphasize the development of emerging athletes, increasing our investment in training new talent, particularly in events where medal counts are currently low, with the aim of achieving breakthroughs in future competitions.

We need to pay particular attention to certain sports. Our analysis indicates that some countries excel in specific events, such as those that completely dominate certain competitions—like the USA (Roque) and SUI (Aeronautics), which have achieved a 100% medal share in their respective events, demonstrating their absolute advantage. Additionally, countries like GBR (Cricket) and USA (Golf) hold a significant proportion of medals in certain events, showcasing their strong competitiveness. In light of this, we propose three recommendations:

- We should study the successful experiences of these countries in specific events, analyzing their training methods, resource allocation, and athlete selection criteria, so that we can implement improvements in our corresponding events.
- Using smaller bathtub if possible
- Decreasing motions during bath
- Using bubble bath additives
- Arranging more faucets of inflow

Furthermore, from the perspective of the correlation between the number of medals and the variety of sports events, our analysis indicates a significant positive correlation (Pearson correlation coefficient of 0.62) between the number of medals won by a country and the number of sports events in which it participates. This suggests that increasing the variety of sports events may enhance a country's medal count in international competitions. Therefore, we recommend that, on one hand, we diversify our sports offerings by encouraging athletes to participate in multiple events, particularly in those where we already have a foundational presence, to improve our overall medal count. On the other hand, we should consider allocating resources to support athletes and coaching teams that engage in multiple sports.

We believe that optimizing the national sports strategy, particularly in terms of diversifying sports offerings, rational resource allocation, and nurturing emerging athletes, will significantly

enhance our performance in international competitions. We look forward to further discussing these recommendations with you and working together to achieve our goals.

Thank you for your attention and support.

Sincerely yours,

Team # 2517063

8 Model Sensitivity Analysis

8.1 Sensitivity Analysis of Input Parameters

We conducted a sensitivity analysis on the input parameters of the model established in response to the first question. Therefore, we conducted sensitivity analyses on the following factors: total number of events, athlete performance (medal count), and the country's historical medal count. In each parameter analysis, we altered only the specified parameter while keeping the other parameters constant.

The Gaussian noise is defined as follows:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (12)$$

Where, x represents the amplitude of the random signal; μ denotes the mean; and σ indicates the standard deviation.

$$\text{Relative error } m_x = \frac{|Pre_{New} - Pre_{Old}|}{Pre_{Old}} \times 100\%.$$

Then, we predicted the total number of events in which the country participates, the performance of its athletes, and the historical medal rates, incorporating Gaussian noise along with the host country effect. Specifically, when Gaussian noise was added to the number of events, we independently observed the prediction results for $x=21,34,65$. When x was set to 21, the relative error reached 8.3%; when x was set to 34, the relative error increased to 14.3%; and when x was set to 65, the relative error rose to 18.4%. Similarly, we tested the performance of athletes and the historical medal rates by adding noise, and the variations in relative error exhibited a certain regularity without any abrupt jumps or anomalous fluctuations. This strongly indicates that the model possesses good adaptability, leading to the conclusion that it is robust in predicting the number of Olympic medals for countries.

9 Model Evaluation

9.1 Strengths

- We employed a multiple linear regression model to predict the medal standings, which is easy to understand and interpret. The coefficients of the model directly reflect the extent of influence that each independent variable has on the dependent variable, facilitating the exploration of new independent variables to meet diverse research needs.
- A significant advantage of our model is its substantial scalability, as it integrates various potential factors into a unified and robust framework, demonstrating effectiveness in predicting future Olympic medals.

- A significant advantage of our model is its substantial scalability, as it integrates various potential factors into a unified and robust framework, demonstrating effectiveness in predicting future Olympic medals.

9.2 Weaknesses

- The presence of high correlation among independent variables may lead to multicollinearity issues, affecting the stability and interpretability of the multiple linear regression model.
- For the predictions of the medal standings for various countries, the prediction intervals provided by our model are sometimes wider than the actual results. This may result in our model's predictions for these countries being less informative.
- Due to the limitations in the collection of sample data, some potential influencing factors may not be adequately analyzed, leading to discrepancies between the predicted and actual results.
- The model makes relatively ideal assumptions about the distribution of the data, which may not hold true in certain cases.

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